

Eukaryote Cells Study Guide

Section B2.2

B2.2.1— Organelles as discrete subunits of cells that are adapted to perform specific functions. *Students should understand that the cell wall, cytoskeleton and cytoplasm are not considered organelles, and that nuclei, vesicles, ribosomes and the plasma membrane are.*

- Define organelle as a discrete structure within a cell that is adapted to perform a specific function.
- List solid organelles and those enclosed by a single membrane and a double membrane.
No membrane: ribosomes, centrioles, nucleolus
Single membrane: vesicles, vacuoles, ERs, golgi, lysosomes, golgi apparatus, rough and smooth ER, vacuoles
Doubles membrane: nucleus, mitochondria, chloroplasts
- Outline the reason why the cell wall, cytoskeleton and cytoplasm are not considered to be organelles.
Cell wall: not in cell
Cytoplasm: not specialized for specific function
Cytoskeleton: not a discrete structure

B2.2.3— Advantages of compartmentalization in the cytoplasm of cells. *Include concentration of metabolites and enzymes and the separation of incompatible biochemical processes. Include lysosomes and phagocytic vacuoles as examples.*

- State the meaning “compartmentalization” within cells
Efficient spaces where multiple activities happen
An organelle creates a membrane bound compartment with controlled conditions inside where multiple activities happen
- List advantages of compartmentalization within cells.
Enzymes + metabolic are in a small space, increasing chance of collision between active site and substrates
Substrates that can damage other cells are isolated in membrane
Conditions stay to specific reactions
Large areas of membrane can become dense with proteins for a specific process
- Outline the benefit of compartmentalization of lysosomes and phagocytic vacuoles in cells.
Because contain high concentration of enzymes, good that lysosomes are bound in membrane since their ph is more acidic (5) where the neutral ph in cytoplasm is 7.2, it can cause damages to the rest of the cell

B2.2.6— Functional benefits of the double membrane of the nucleus. *Include the need for pores in the nuclear membrane and for the nucleus membrane to break into vesicles during mitosis and meiosis. (this is covered in more detail in the cell division unit)*

- Describe the structure and function of the pores in the nuclear membrane.
Structure of pores: have proteins that can control what goes in and out, formed using integral proteins

Function: to allow proteins from cytoplasm to nucleus then vice versa, nm breaks down to allow chromosomes to be moved to poles then form around new group of chromosomes

B2.2.7— Structure and function of free ribosomes and of the rough endoplasmic reticulum.

Contrast the synthesis by free ribosomes of proteins for retention in the cell with synthesis by membrane- bound ribosomes on the rough endoplasmic reticulum of proteins for transport within the cell and secretion.

- Outline the structure of ribosomes.

Made from dozens of proteins arranged on a scaffold of ribosomal RNA

Two subunits come together

- Compare the structure and function of free ribosomes and those bound to the rough endoplasmic reticulum.

Free ribosomes:

Structure: floating in cytoplasm

Function: synthesizing polypeptides used within the cell

Bound ribosomes:

Structure: attached to RER

Function: synthesizing polypeptides that are secreted from cell or become integral proteins in cell membrane

- Compare the ultimate destination(s) of proteins synthesized at free ribosomes to those synthesized at ribosomes bound to the rough endoplasmic reticulum.

free ribosomes destinations: cytoplasm, nucleus, mitochondria or stay in ribosome

Bound ribosomes destinations: golgi apparatus

B2.2.8— Structure and function of the Golgi apparatus. *Limit to the roles of the Golgi apparatus in processing and secretion of protein.*

- Outline the structure and function of the Golgi apparatus.

Structure: flattened membrane-enclosed sacs

Function: polypeptides modified into functional state

- Outline the general flow of proteins to, through and from the Golgi apparatus.

Polypeptides synthesized by ribosomes bound to ER transported in vesicles to Golgi, vesicles fuse with Golgi on cis face, polypeptides then transported through Golgi and exit via vesicles budded from trans face

B2.2.9— Structure and function of vesicles in cells. *Include the role of clathrin in the formation of vesicles.*

- Describe the formation of vesicles in cells from phospholipid membranes, including the role of clathrin.

Membrane bulges and pinches off with clathrin “coat” that helps phospholipids create a round shape

- List functions of vesicles in cells.

Transport vesicles: transports proteins and lipids

Secretory vesicles: transport proteins and lipids from inside from cell to plasma membrane

- Outline the consequence of vesicle fusion.

Is the merging of a vesicle with another organelle or part of cell membrane

Adds phospholipids to target structure, making it bigger

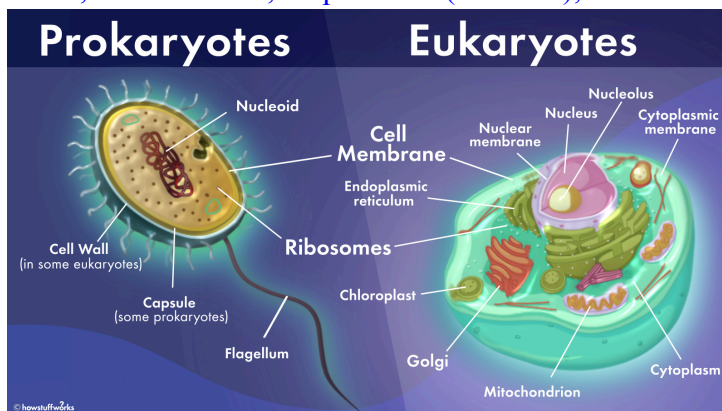
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A2.2.6 —Eukaryote cell structure. *Students should be familiar with features common to eukaryote cells: a plasma membrane enclosing a compartmentalized cytoplasm with 80S ribosomes; a nucleus with chromosomes made of DNA bound to histones, contained in a double membrane with pores; membrane-bound cytoplasmic organelles including mitochondria, endoplasmic reticulum, Golgi apparatus and a variety of vesicles or vacuoles including lysosomes; and a cytoskeleton of microtubules and microfilaments.*

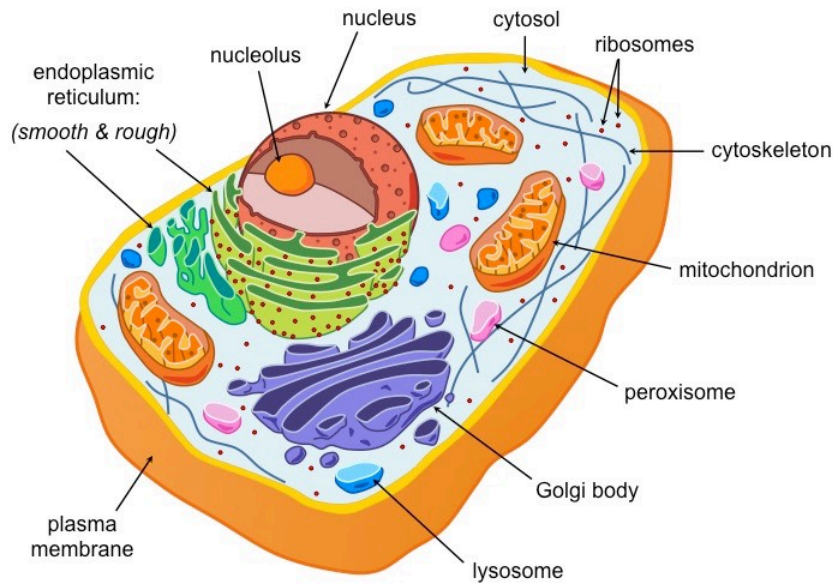
- Compare and contrast prokaryotic and eukaryotic cell structure.

Prokaryotic: smaller, no membrane-bound organelles, division by binary fission, cell wall (not cellulose), DNA is nucleoid (no nuclear membrane), circular and naked DNA, 70s ribosome, have plasmids, unicellular

Eukaryotes: bigger, membrane-bound organelles, division by binary fission/mitosis/meiosis, cell wall of cellulose/chitin, DNA in nucleus, linear DNA not naked, 80s ribosome, no plasmids (for most), unicellular/multicellular



- Label a diagram of a eukaryotic cell.



- Outline the function of the following structures in the eukaryotic cell: plasma membrane, cytoplasm, 80s ribosomes, nucleus, mitochondria, chloroplast, endoplasmic reticulum, Golgi apparatus, vesicles, vacuoles, lysosomes, microtubules, and cytoskeleton

Plasma membrane	Membrane that surrounds and protects cell
cytoplasm	Everything in between the nuclear membrane and plasma membrane
80s ribosome	Catalyzes synthesis of polypeptides
Nucleus	Holds' cells chromosomes (DNA) and regulates genes
Mitochondria	Carry out aerobic respiration, converts nutrients to energy, ATP synthesis
chloroplasts	Adapted for photosynthesis, captures light energy to use with H ₂ O and CO ₂ for glucose
Endoplasmic reticulum (ER)	Roles in synthesizing and transport of polypeptides
Golgi apparatus	Modifies polypeptide into functional state
vesicles	Membrane bound sacs that contain and transport materials
Vacuoles	Store water and other dissolved materials

lysosomes	Contain digestive enzymes to break down ingested food in vesicles or break down organelles or whole cell
microtubules	Proteins that form cytoskeleton, moves chromosomes during cell division (mitosis & meiosis)
Cytoskeleton	Helps maintain shape, organizes cell parts and enables cells to move and divide

A2.2.8— Differences in eukaryotic cell structure between animals, fungi and plants.

Include presence and composition of cell walls, differences in size and function of vacuoles, presence of chloroplasts and other plastids, and presence of centrioles, cilia and flagella.

- Compare and contrast the structures of plant, animal and fungal cells with reference to cell walls, vacuoles, chloroplasts, centrioles, cilia and flagella.

	Plant	Animal	Fungal
Cell wall	Composed of cellulose	none	Composed of chitin + other
vacuoles	Large and permanent	Small	Large and permanent
chloroplast	present	Not present	Not present
centrioles	Present in gametes of moss and ferns Absent in conifer + flowering plants	Used to arrange protein fibers during cell division	Absent Other than some exceptions where sperm for male
Cilia	Present: male gametes of moss and ferns	Present including male gamete	Absent Except for some exceptions for male sperm
Flagella	Present: male gametes of moss and ferns	Present including male gamete	Absent Except for some exceptions for male sperm

A2.2.9— Atypical cell structure in eukaryotes. *Use numbers of nuclei to illustrate one type of atypical cell structure in aseptate fungal hyphae, skeletal muscle, red blood cells and phloem sieve tube elements.*

- Describe features of skeletal muscle fibers that make them an atypical cell.
Many nuclei
One big one where multiple cells fused together, not individual
- Describe features of aseptate fungal hyphae that make them an atypical cell.
Many nuclei
One big one where multiple cells fused together, not individual
- Describe features of red blood cells that make them an atypical cell.
No nucleus
Eukaryotic with no nucleus or mitochondria
- Describe features of phloem sieve tube elements that make them an atypical cell.
No nucleus
Cell without organelles
- Compare the number of nuclei in aseptate fungal hyphae, skeletal muscle, red blood cells and phloem sieve tube elements.
Red blood cell + phloem sieve tube = no nucleus
Aseptate fungal hyphae + skeletal muscle fiber = many nuclei

A2.2.10— Cell types and cell structures viewed in light and electron micrographs.

Application of skills: Students should be able to identify cells in light and electron micrographs as prokaryotes, plant or animal. In electron micrographs, students should be able to identify these structures: nucleoid region, prokaryotic cell wall, nucleus, mitochondrion, chloroplast, sap vacuole, Golgi apparatus, rough and smooth endoplasmic reticulum, chromosomes, ribosomes, cell wall, plasma membrane and microvilli. (Also in the origin of cells unit)

- Recognize features and identify structures in micrographs of eukaryotic cells (inclusive of the plasma membrane, nucleus, mitochondrion, chloroplast, vacuole, rough and smooth endoplasmic reticulum, Golgi apparatus, secretory vesicle, ribosomes, cell wall, and microvilli)

A2.2.11— Drawing and annotation based on electron micrographs. *Application of skills: Students should be able to draw and annotate diagrams of organelles (nucleus, mitochondria, chloroplasts, sap vacuole, Golgi apparatus, rough and smooth endoplasmic reticulum and chromosomes) as well as other cell structures (cell wall, plasma membrane, secretory vesicles and microvilli) shown in electron micrographs. Students are required to include the functions in their annotations. (Also in the origin of cells unit)*

- Given a micrograph, draw and label the ultrastructure of a eukaryotic cell.

A2.2.12— Origin of eukaryotic cells by endosymbiosis. *Evidence suggests that all eukaryotes evolved from a common unicellular ancestor that had a nucleus and reproduced sexually. Mitochondria then evolved by endosymbiosis. In some eukaryotes, chloroplasts subsequently also had an endosymbiotic origin. Evidence should include the presence in mitochondria and chloroplasts of 70S ribosomes, naked circular DNA and the ability to replicate.*

- Explain the origin of mitochondria and chloroplast with reference to the endosymbiosis.
Eukaryotic cells engulfed aerobic bacteria but did not digest, lead to mitochondria
Eukaryotic cells engulfed photosynthetic bacteria but did not digest, lead to chloroplasts

- Describe the genetic, structural and behavioral evidence for the endosymbiotic theory.
Structural: mit. + chloro have similar shape and size w double membrane
Genetic: mit. + chloro: circular DNA, common DNA sequence
Functional: mit. + chloro. Move independently within, reproduce independently, inhibited by antibiotics like prokaryotic